

Singh Sanskar Surendrasingh

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Research Interests

I work on efficient and reliable information retrieval, spanning vector search, multimodal retrieval and retrieval-grounded generation. I study how embedding geometry and query characteristics shape retrieval behaviour, to build systems that adapt to each query and maintain quality under compute and memory budgets.

Education

Indian Institute of Technology (IIT) Kharagpur

CGPA: 8.46 / 10

Master of Technology in Wireless Communication & Networks

Jun 2023 – May 2025

- Master's Thesis: *Facial Recognition and Verification using a Vision-Transformer Siamese Network* (advised by Prof. Saumik Bhattacharya & Prof. Suva Sekhar Das).

Gujarat Technological University

CGPA: 7.83 / 10

Bachelor of Engineering in Electrical Engineering

Jun 2019 – May 2023

Publications

[1] Scalability Analysis of Cross-Modal Text-to-Image Pipelines

Sanskar S. Singh, Ziv Barretto, Bharati Khanijo, Aditya Sahu, Maya Ramanath

CIKM 2026 • Full Paper • Accepted

Systems-level study of single-index, rerank and ensemble text-to-image pipelines on 7M LAION images, measuring accuracy, latency, energy and storage. **O-IVFPQ is the best single index**: it finds 84% of the exact top-100 results (HNSW: 91%) with 10× less storage and 1.8× lower latency. Rerankers lose only 2–5% R@1 when the index finds 70–80% of them, but **dominate end-to-end cost** (1–4.5 s vs. <60 ms per query).

[2] Retrieval-Stage Interactions in Cross-Modal Text-to-Image Retrieval Pipelines

Sanskar S. Singh, Ziv Barretto, Bharati Khanijo, Aditya Sahu, Maya Ramanath

CIKM 2026 • Short Paper • Accepted

Showed that **reranking gains depend on first-stage quality**: on MS-COCO, BLIP-2 lifts CLIP's R@1 from 0.38 to 0.65, while a weak reranker cuts UniLR's from 0.55 to 0.39. RRF ensemble gains come from **complementary query coverage**, not raw accuracy.

[3] Fidelity Synopses: Pre-Execution Plan Selection for Approximate Vector Search

Adithya R, Sanskar S. Singh, Maya Ramanath

VLDB 2027 • Full Paper • Submitted

Showed that vector databases, which typically run every query on one fixed ANN index setting, could meet the same accuracy target at **about half the cost (up to 7× less)** if each query got its cheapest sufficient setting. Proposed the **spatial fidelity map**, a compact offline synopsis that predicts, from one landmark lookup, what share of the true top-100 neighbours each setting will return, with **12–38% lower error** than a global-average baseline across 4 embedding models and 4 search directions.

[4] SetCoRe: Set Composition for Retrieval of Educational Explanations

Riti Verma, Sanskar S. Singh, Maya Ramanath

ECIR 2027 • Full Paper • Submitted

LLM-free retriever that selects complementary textbook resources by predicted set composition and concept coverage. It reaches F1 0.415, above the strongest deployable non-LLM baseline (0.395) and close to a Llama-3.1-70B-based selector (0.451), while running **300× faster**.

[5] Set2Story: Retrieval-Grounded Storyboard Construction for Educational Explanation

Aryan Ahmed, Sanskar S. Singh, Maya Ramanath

ECIR 2027 • Full Paper • Submitted

Defined grounded storyboard construction: turning retrieved resources into a timed video plan whose segments carry teaching roles (hook, definition, example, ...) Our planner (<0.6M-param role predictor, NLI-verified narration) beats the strongest baseline, a fixed-template LLM pipeline, on **role choice (F1 0.808 vs. 0.713)** and **role order (0.757 vs. 0.666)**, and raises **supported narration claims from 31% to 99.7%** over free-form LLM generation.

[6] SARCH: Multimodal Search for Archaeological Archives

N. Sinha, B. Khanijo, Sanskar S. Singh, et al.

CIKM'25 MMGenSR Workshop, Seoul • Published

Hybrid search over 296 scanned archaeology documents (63K+ pages), fusing MiniLM, CLIP and TAPAS embeddings with BM25 via reciprocal-rank fusion.

Research Experience

IIT Delhi (Advisor: Prof. Maya Ramanath)

New Delhi, India

Project Scientist – Multimodal Information Retrieval, DSIR Lab

May 2025 – Present

- Research spanning **information retrieval, multimodal retrieval, representation learning, NLP** and **optimisation**, including vector databases and ANN search, energy-efficient retrieval, reranking and rank fusion, educational retrieval, and grounded LLM generation. Team lead in the lab, guiding research students on research work and paper writing; Teaching Assistant for COL2010: Data Science.

Projects

IIT Kharagpur

One-Shot Facial Verification with a Siamese Network (Master's Thesis)

Jun 2024 – May 2025

- Trained a **Vision-Transformer Siamese model** on Celeb-DF v2 (31K+ images) for one-shot recognition and verification, using balanced (anchor, positive, negative) triplets, a metric-learning setup that mirrors learned similarity in dense retrieval.
- Achieved **94.83% recognition accuracy** and **91.06% verification accuracy** (98.44% precision, 85.77% recall).

Industry Experience

Eternal Robotics

Ahmedabad, India

Computer Vision Intern

Feb 2025 – May 2025

- Integrated and deployed a YOLO-based quality-inspection system for car assembly lines, and built an OCR proof-of-concept on vehicle stickers with Qwen2.5-VL-Instruct (**94.74% accuracy**); contributed to a CI/CD pipeline for model deployment.

Unizoy

Surat, India

Machine Learning Engineering Intern

May 2024 – Jul 2024

- Developed a real-time pose-estimation and feedback system (Mediapipe) covering 10+ exercises for remote physiotherapy, reaching **93% accuracy** on the biceps-curl model.

Teaching Experience

Department of Computer Science and Engineering, IIT Delhi

Delhi, India

Teaching Assistant – COL2010: Data Science

Sep 2026 – Present

- Designed course assignments and evaluated submissions, providing feedback to assess understanding and reinforce key concepts.

Awards

Ideathon, SSGEC Bhavnagar

Bhavnagar, India

Team Leader – 1st Place

Feb 2022

- Won 1st place for proposing a renewable energy-storage solution that harnesses kinetic energy from flowing river water and converts it to stored potential energy.

Skills

Languages: Python, C

ML / Deep Learning: PyTorch, TensorFlow, Keras, Hugging Face Transformers

Retrieval & IR: FAISS (KNN / HNSW / IVFPQ / O-IVFPQ), Apache Solr, BM25

Multimodal Models: CLIP, OpenCLIP, SigLIP2, Jina-v4, FLAVA, UniLR, BLIP-2, Qwen3-VL, BGE

Tools & Libraries: Git, FastAPI, Flask, Streamlit, NumPy, pandas, Matplotlib, OpenCV, Linux

Spoken Languages: English (full professional), Hindi (native)